ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. CSE ENGINEERING — SEMESTER 6

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain phases of a Compiler with a sample statement: position = initial + rate * 60. [14 Marks]

[OR]

Q.1 (Alt) Explain Lexical Analyzer role, regular expressions, transition diagrams, and Lex tool. [14 Marks]

--- UNIT II ---

Q.2 Construct LL(1) parsing table for given grammar. Check for ambiguity and left recursion. [14 Marks]

[OR]

Q.2 (Alt) Construct SLR(1) and LR(1) parsing tables with canonical collections of LR items. [14 Marks]

--- UNIT III ---

Q.3 Explain Syntax-Directed Translation (SDT) and S-attributed vs L-attributed definitions. [14 Marks]

[OR]

Q.3 (Alt) Generate Three-Address Code, Quadruples, Triples, and Abstract Syntax Trees. [14 Marks]

--- UNIT IV ---

Q.4 Explain Run-Time Storage Administration: Activation Records, Stack allocation, and Heap. [14 Marks]

[OR]

Q.4 (Alt) Describe Symbol Table organization techniques and Scope management. [14 Marks]

--- UNIT V ---

Q.5 Explain Principal Sources of Code Optimization: Common Subexpression elimination, Loop unrolling. [14 Marks]

[OR]

Q.5 (Alt) Explain Target Code Generation, Register allocation, and Basic Blocks DAG representation. [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****